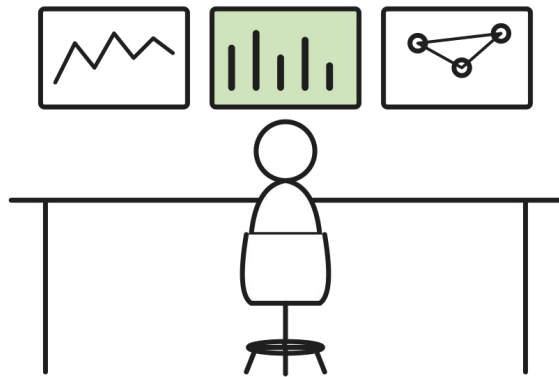


Doing ML Research from the Margins



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CMU LTI · AWS Bedrock · ML Collective Africa

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Wits University · ML Collective Africa

mlcollective.org

OUTLINE OF TODAY'S SESSION

- 01 **Framing**
- 02 **Finding mentors**
- 03 **Scoping research**
- 04 **Reading the literature**
- 05 **Top venue positioning**

01

What the path looks
like in retrospect

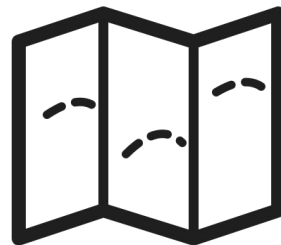
The barrier is high but the people who broke through
all made the same identifiable moves.



Luck in research is not randomly distributed.

Privilege compounds: resources → results → more resources. That's cumulative advantage, and it's why the playing field isn't level.

But the people who break through from constrained contexts make a **small number of identifiable moves**.



Different starting points, same navigable territory

Steven K.

From: FUNAAB undergrad, no lab

To: CMU PhD, science @FAANG, NeurIPS

Ifihan O.

From: UNILAG undergrad

To: Wits University

Chisom

From: Nigerian undergrad, MLC member

To: Cambridge MSc

Zaynab A.

From: UNILAG undergrad

To: Max Planck Inst., EPFL (still an undergrad!)

Oreva A.

From: AI Saturdays Lagos

To: UW PhD, Google, Apple, Top NLP publications

Yusuf

From: UNILAG undergrad

To: CMU Africa / NTU Singapore

FOUR ESSENTIAL MOVES



Find community early

Join communities (like ML Collective, IndabaX, AI Saturdays, etc) to connect with peers and mentors early, even before you have formal credentials.



Make yourself a worthwhile bet

Deliver a small, concrete contribution first. Reaching out with an actual piece of work to show dramatically increases your response rate.



Scope first work carefully

Your initial paper doesn't need to be highly ambitious. Keep the scope manageable so you can focus on simply getting it done.



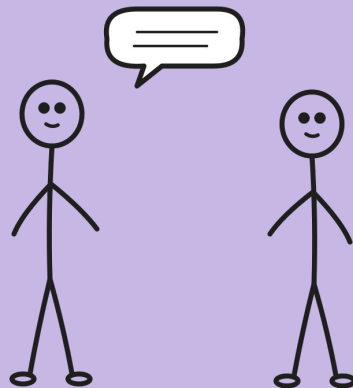
Build momentum from small wins

Progress from workshops to active feedback to stronger submissions. Each small milestone builds confidence and eases the next step.

02

Finding and approaching mentors

How do you find and approach prospective mentors?



The mentor-finding problem is a funnel



STAGE 1

Find the right people

Not the most famous but the most accessible.
Reachable researchers in your specific subfield.

STAGE 2

Make contact that gets a
response

Most outreach fails here. The difference is
specificity, not sincerity.



most outreach fails here

STAGE 3

Build the relationship

Convert a response into an ongoing
collaboration. Show momentum. Don't
disappear.

WHERE TO FIND ACCESSIBLE SENIOR RESEARCHERS



Open-science communities

Communities like ML Collective host researchers who actively welcome outreach, turning cold emails into warm introductions.



Workshop author lists

Authors presenting at recent specialized workshops are highly active in your specific domain right now.



Paper footnotes & acknowledgments

Look for 'The authors thank X.' These individuals are often PhD students eager to collaborate, with direct contact info listed.



X / Twitter

Active posters and repliers are signaling their availability. Build rapport by engaging publicly before sending a direct message.



IndabaX & local events

Connecting in person at localized meetups completely transforms the dynamic, converting cold outreach into a warm connection.



Google Scholar & lab websites

Target mid-career researchers with 5 to 15 papers. They are established enough to offer valuable mentorship, yet accessible enough to reply.

ANATOMY OF A COLD EMAIL



The hook

Name one specific thing from their work. Generic compliments are invisible.

Your context

Two or three sentences. Who you are, what you've built. Not your life story.

The ask

Small, direct, answerable. A specific question — not 'be my mentor.'

The exit ramp

'No worries if you don't have bandwidth.' Be confident, not desperate.

What not to do

- × Generic email that could go to anyone
- × Asking for mentorship upfront
- × No concrete question
- × Your full life story
- × Mass-emailing one template — they talk to each other

EXAMPLE

Subject: Question about your low-resource NER results

Hi Dr. Adeyemi,

I read your EMNLP paper on augmentation for low-resource NER — the result that back-translation hurt Yoruba performance surprised me.

I'm an independent researcher with X working on NER for Nigerian Pidgin, and I've reproduced your baseline on my own dataset.

Did the degradation hold when the augmentation model was trained monolingually? I couldn't find it in the appendix.

No worries if you don't have bandwidth to reply — the paper was genuinely useful either way.

← The hook

← Context

← The ask

← The exit

TIMING AND FOLLOW-UP



When to reach out

Best	Right after a workshop acceptance
Good	After meeting them at a conference
Okay	With a concrete, well-formed question
Bad	Before you have any work done

After you get a response

- Short update every 2–3 weeks — a paragraph, not an essay
- Show momentum: what you ran, what you found, what's next
- Don't vanish, then reappear asking for a rec letter
- One polite follow-up after a week. Then move on.

03

Scoping tractable research

You don't need a massive cluster to do good research.



The three-property tractability filter

1

Requires a low-cost signal

Avoid projects needing closed datasets or massive compute (like 8 A100s). Aim for problems where open weights and a Colab notebook provide 80% of the necessary signal.

2

Ensure a completable, publishable unit

Identify the smallest honest, novel version of your contribution. A polished workshop paper is far more valuable than an unfinished full-conference submission.

3

Remains relevant to the frontier

Ensure the problem is one top venues actually care about. Most researchers start here, but the common mistake is overlooking requirements (1) and (2).

All three criteria must be met simultaneously.

Will it need massive compute? Ask before you start

You can estimate the bill before writing a line of code:

- 1 **Training from scratch, or adapting?** Fine-tuning, LoRA, probing, prompting, evaluating (cheap). Pretraining (could be expensive)
- 2 **Does the model fit in 16GB?** Open weights up to ~8B, quantized, run on a free Colab or Kaggle GPU.
- 3 **What hardware do your baseline papers report?** If their experiments took 64 GPUs for a week, your reproduction won't be cheap either. Rescope.
- 4 **Can one experiment finish overnight on Colab?** If a single run takes days on a free GPU, the iteration loop is dead before you start.
- 5 **Is the data open and small?** Closed data or terabytes both kill the project at step one.

Ideas are free, compute isn't

Three ways in

1 Build on paid-for artifacts

Open weights and checkpoints someone else spent millions training. Fine-tune, probe, merge, evaluate.

2 Study small models

Do the science at 1B, argue the findings transfer. Small-model science is now a genre, not a compromise.

3 Own a neglected niche

The directions big labs don't prioritize are wide open and often exactly where your context is the advantage.

Ideas, data, analysis. Pick the bottleneck you can win.

Directions wide open right now

- Low-resource & multilingual NLP
- Efficient inference, compression & edge
- Mechanistic interpretability
- Test-time compute & reasoning
- Evaluation & benchmarking
- Data-centric work & small-model science
- Model merging & souping

Your local context is an asset, not a liability

The reframe

Top labs have no incentive to work on low-resource languages, African clinical datasets, or non-English evaluation gaps. Problems invisible to them are exactly where the gaps are.

The frontier is not one place. It has gaps you can see that they can't.

Worked examples

Afri-MCQA (ACL 2026)

Multilingual QA benchmark for African languages — invisible to labs with no African language speakers.

Masakhane

African researchers on African NLP problems, publishing at ACL, EMNLP, and beyond.

Local data collection

Hand-collected sign language data at Nigerian schools → National AI Champion recognition.

Don't wait until you have results to start writing.



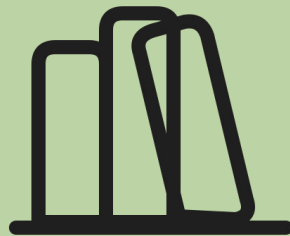
Writing forces you to be clear about what you don't understand.

It crystallises the contribution. And it keeps your momentum alive.

04

Reading the literature efficiently

How to get up to speed and get noticed while doing it.



THE THREE-PASS METHOD ACROSSING TO KESHAV

Pass 1 5–10 min

Is it worth your time?

- Title, abstract, intro
- Section headings
- Conclusions

You can name the category, context, and contribution — or you stop here.

Pass 2 ~1 hour

Grasp the content

- Read carefully, skip proofs
- Study every figure
- Note unread references

You can summarise the main thrust, with evidence, to someone else.

Pass 3 4–5 hours

Deep mechanics

- Mentally re-implement it
- Challenge every assumption
- Note hidden failings

Only for papers central to your research. You can rebuild it from memory.

HOW TO MOVE FROM ZERO TO FRONTIER-AWARE IN A WEEK

1

Find 3–5 recent papers

Google Scholar or arXiv, filtered to the last year or two.

2

Trace shared citations

Papers cited by several of them are the canon.

3

Spot the key researchers

Recurring names = who the community trusts.

4

Note the top venues

Where the best people publish is where to aim.

5

Iterate

Missed a key paper? Get it, repeat. 2–3 rounds.

ENGAGING PUBLICLY BUILDS REPUTATION



Share your reading notes

Summaries, mind maps, annotated PDFs. People find them and reach out. Networking without the small talk.



Implement a paper, share the repo

A working re-implementation is the strongest possible signal that you know what you're doing.



Ask questions in the open

Confused by a result? Ask on X or in community Discords, and tag the authors.

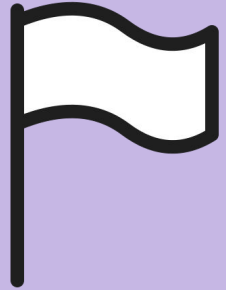


Warm up your future outreach

Reply to a researcher's posts before you email them.

05

Writing and positioning for top venues



Understand the game of peer review.



How peer review actually works

The idealized version

Scholars with leisure time, carefully poring over the nuances of your work, weighing every point before a verdict.

The reality

A crowded marketplace. Reviewers read 6–10 papers in a short window. Not your opponents — but not your advocates either.

The five-minute test: can a reviewer reconstruct your contribution in five minutes?

If not, the paper is at risk regardless of the work underneath.

Why we self-reject — and default to lower venues

It is rarely about the quality of the work. It is almost always about navigation.

Not knowing venue fit

Sending a medical imaging paper to ACL. Without labmates to calibrate against, the venue hierarchy is invisible.

No feedback loop

Labs workshop papers internally before submitting. Independent researchers submit cold.

Imposter syndrome

Genuinely novel work gets framed as 'preliminary' because nobody is there to say 'this is actually good.'

Risk aversion

A workshop rejection feels safer. But workshops rarely lead anywhere — the calculus doesn't hold.

HOW FRAMING CHANGES THE RECEPTION

Workshop framing

“We explore whether technique X can be applied to low-resource language Y and present preliminary results.”

Signals: exploratory, unsure, not ready.

Cycles through workshops without ever becoming a full paper.

Conference framing

“We demonstrate that technique X, applied to low-resource language Y, achieves Z improvement — the first result of this kind.”

Signals: confident, specific, measurable.

The same experiments, but presented as a clear contribution rather than a tentative exploration

Write with command



Active voice

“We show that...” — not “It could be argued that...”. Own your claims.

Concrete and measurable

“14% improvement on X” beats “promising results” every time.

Contribution up front

State it plainly in the abstract and again on page one. Don't make reviewers dig.

Pass the 5-minute test

Title + abstract + figures should carry the whole story on their own.

STRATEGIC PLANNING FOR CONFERENCE CYCLES



AI Deadlines · aideadlin.es

Countdown timers for every major ML venue, filterable by subfield. Bookmark it.



CCF DDL · ccfdl.com

Broader CS conference tracker with rankings and historical dates.



WikiCFP · wikicfp.com

Calls for papers across every field, including workshops. Good for smaller venues.



OpenReview venue pages · openreview.net/venues

Each venue's full timeline: abstract, submission, rebuttal, decision.



Papers With Code · paperswithcode.co/ai-deadlines

Upcoming submission deadlines for major AI conferences, counting down.



Follow the venues on X

@NeurIPSConf, @iclr_conf, @aclmeeting announce extensions and changes first.

OPENREVIEW: YOUR FREE ADVISOR

How to use it

1. Search a venue in your area (ICLR, NeurIPS, ACL)
2. Filter to recent accepted papers
3. Find one adjacent to your work
4. Read every review — including the low scores
5. Read the rebuttals
6. Notice what reviewers consistently reward

What to look for

Clarity of contribution — How accepted papers nail it in the abstract.

Expected baselines — Which comparisons your subfield demands.

Framing novelty — The language patterns are learnable.

Rebuttals that flip decisions — Some do, some don't. The difference is instructive.

WHAT REVIEWERS LOOK FOR



Novelty

A genuinely new idea — not incremental tuning.

Significance

Does it change how people think, or enable something new?

Soundness

Claims supported, baselines fair, assumptions stated.

Clarity

Contribution reconstructable in five minutes — or the paper is at risk.

Related work

Understand your context. A missed citation raises a red flag, whereas a thorough related work section establishes instant credibility.

Rebuttals: The Invisible Phase That Can Save Your Paper

After reviews, you get a brief ~500-word window to respond before the final decision.

Address Two Audiences

Tailor your response to both stakeholders: the active reviewers and the Area Chair who holds the final decision-making power.

Lead with Your Answers

Directly quote each concern and answer it immediately. Reviewers are busy and will likely skim your response.

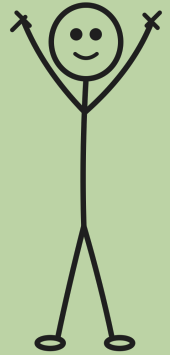
Choose Your Battles

Focus strictly on correcting clear factual errors, addressing misunderstandings, and making claims you can robustly back up with evidence.

Avoid Being Defensive

Aim to clarify and convince rather than trying to win an argument. Acknowledging valid critiques often disarms defensive reviewers.

The goal was never to inspire.
It was to leave you with a clearer
map than we had.



The path exists. It's navigable. And now it's a little more legible.

RESOURCES

Finding mentors

MLC Africa Mentor Outreach Templates

mlcollective.org/wiki/mlc-nigeria/mentor-outreach-email-templates

Scoping research

Simon Peyton Jones — How to Write a Great Research Paper

simon.peytonjones.org/great-research-paper

Andrej Karpathy — A Survival Guide to a PhD

karpathy.github.io/2016/09/07/phd

Reading the literature

Keshav — How to Read a Paper (3-pass method)

web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf

Top venue positioning

OpenReview — read live reviewer comments in your area

openreview.net

Colin Raffel — Reviewing Criteria

colinraffel.com/blog/reviewing-criteria.html

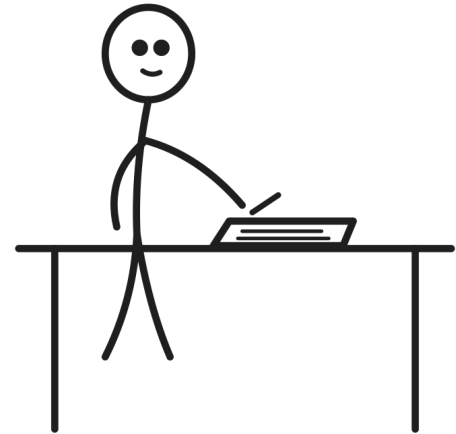
Parikh, Batra & Lee — How We Write Rebuttals

deviparikh.medium.com/how-we-write-rebuttals-dc84742fece1

Take-home exercise

Draft your outreach
email this week.

Pick one researcher whose work is adjacent to yours.



Questions & Discussion

We're also around after the session. Come find us.

Steven Kolawole

CMU LTI · AWS Bedrock · ML Collective Africa

Ifihanagbara Olushey

Wits University · ML Collective Africa

mlcollective.org

